

## Year 2 Quantum Mechanics - Example Sheet 1

1. Show by substitution that the wave function

$$\Psi(x,t) = Ae^{i(kx - \omega t)}$$

is a solution of the time-dependent Schrödinger equation when the potential  $V$  is not a function of position.

Similarly show that the wavefunction

$$\Psi(x,t) = A \sin(kx - \omega t)$$

is not.

2. Show that if  $\psi_1(x,t)$  and  $\psi_2(x,t)$  are both solutions of the Schrödinger equation, then so is  $\psi(x,t) = C_1\psi_1(x,t) + C_2\psi_2(x,t)$ , where  $C_1$  and  $C_2$  are arbitrary constants.
3. A proton with kinetic energy of precisely 10 MeV moves along the  $x$ -axis. Calculate its (unnormalised) wavefunction and show that the probability of finding the proton at any point on the  $x$ -axis is independent of the value of  $x$ . (Assume  $V=0$ .)
4. The one-dimensional spatial wavefunction describing a particle is given by:

$$\psi(x) = \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{L}\right) \quad -L/2 \leq x \leq L/2$$

$$\psi(x) = 0 \quad x < -L/2 \quad \text{and} \quad x > L/2.$$

Calculate the probability of finding the particle in the range  $-\infty < x < \infty$  and comment on the result.

Calculate the probability that the particle is found in the region  $-L/4 \leq x \leq L/4$ .

Calculate  $\langle x \rangle$ ,  $\langle p \rangle$ ,  $\langle x^2 \rangle$ ,  $\langle p^2 \rangle$ .

5. The first excited state of a one-dimensional simple harmonic oscillator potential is described by the spatial wavefunction

$$\psi(x) = A \left( \frac{x}{a} \right) e^{-x^2/2a^2}$$

Normalise this wavefunction and roughly sketch the shape of the spatial probability.

You will need the result  $\int_{-\infty}^{\infty} x^2 e^{-Kx^2} dx = \pi^{1/2} / (2K^{3/2})$ .